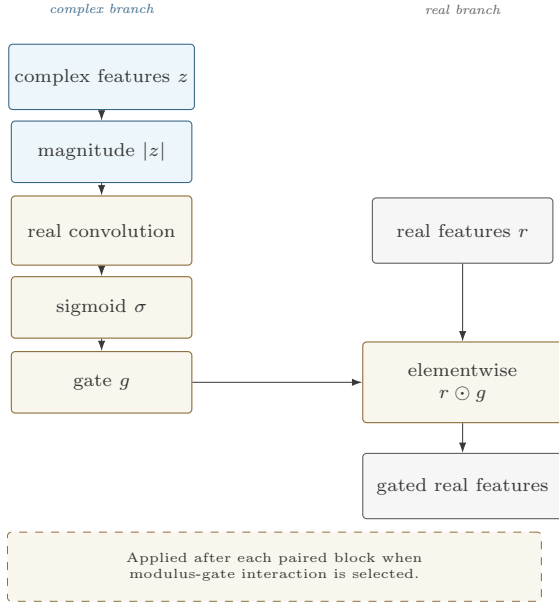


A. Modulus cross-stream gate



B. Complex-branch holographic attention

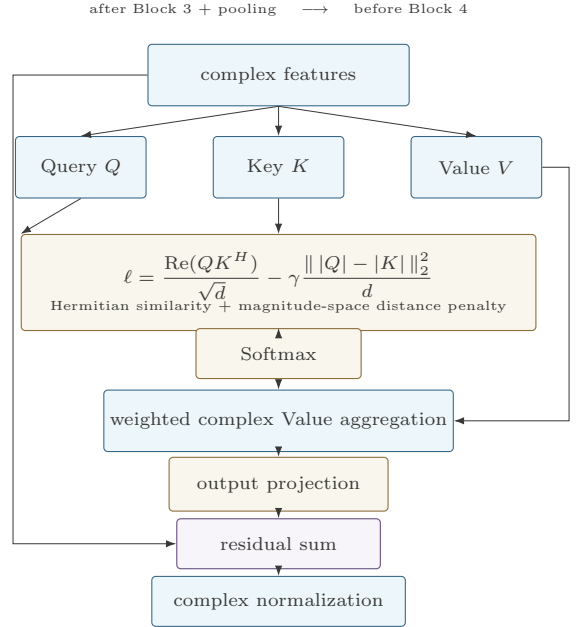


Figure 5. Implemented interaction mechanisms. The modulus gate derives a sigmoid gate from complex-feature magnitudes and applies it multiplicatively to the corresponding real-stream features after each paired block. Holographic attention operates only on the complex branch after the third pooled stage, combining Hermitian query–key similarity with a magnitude-distance penalty before complex value aggregation.